CWI
Centrum Wiskunde & Informatica

AMS DB

Dutch Data Systems

Peter Boncz
CWI + VU

a perspective from CWI

CWI
Centrum Wiskunde & Informatica

Weaviate

ClickHouse

databricks

Polars

elasticsearch

DuckDB Labs

MotherDuck

Outline

- **Intro**
  - Data systems research @ CWI DA
- **Past**
  - MonetDB & VectorWise + impact
- **Present**
  - DuckDB (“in-process analytics”)
- **Future**
  - Research into Data Systems

Data Systems Research

- data systems: of the most **complex software**
  - 10-100Ks of source code, takes 5-10y and 100sFTE
  - compression, data structures, algorithms, optimization, machine learning, compilers, operating systems, hardware co-design

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  - Turing Awards: Jim Gray, Michael Stonebraker

Data Systems Research at CWI

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Martin Kersten

1953-2022

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  - Learned Data Formats
  - SQL:2023 (Property Graph Queries)
  - Responsible Decentralized Data Architectures

Main Memory Column Stores

- Hardware Trends
  - Databases ⇒ Big Data
  - big memories ⇔ memory wall

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Data Distilleries
driving e-business personalization

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- MonetDB (1994-) *Martin Kersten, Niels Nes, Sjoerd Mullender, ..*
  - column-at-a-time processing
  - CPU & *cache-conscious* algorithms

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monetdb

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monetdb

Impact ⇒ SAP HANA (2010) + column store revolution (2005-)

Vectorized Query Processing

- MonetDB scalability..
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  - VectorWise system (2005-) *Marcin Zukowski, Sándor Héman, ..*
  - scaling beyond RAM (“cooperative scans”)

vectorwise

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vectorwise

Impact ⇒ Google Dremel (F1, BigQuery), Apache Drill, Apache Impala, Dremio, CockroachDB, Databricks Delta Engine, Microsoft SQLserver, IBM DB2 Blu, MySQL Heatwave (Oracle), ClickHouse, Snowflake, …, …, …,

VectorWise Data Format

Compressed columnar storage

- **vector**-at-a-time **decoding** algorithms
  - decompress small vectors into **CPU cache**
  - **Run-Length, Zero Truncation (FOR), Delta coding, Bit-Packing**
  - remove outliers (“patching”)

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Compressed columnar storage

- vector-at-a-time decoding algorithms
  - decompress small vectors into CPU cache
  - Run-Length, Zero Truncation (FOR), Delta coding, Bit-Packing
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- **Skippable Row-groups**
  - **= large table chunks** with inside columnar storage (PAX)
  - **Min/Max** for each column in header ⇒ exploit natural orderings

VectorWise Data Format

Compressed columnar storage

- Vectorizable encoding algorithms
  - decompress small vectors into CPU L1 cache
  - Run-Length, Zero Truncation (FOR), Delta coding, Bit-Packing
  - remove outliers (“patching”)
- Skippable Row-groups
  - Columnar storage per large table chunk
  - Min/Max for each column in header ⇒ exploit natural orderings

Impact ⇒ Lucene, Dremel, ORC, **Parquet** (yottabytes)

VectorWise ⇒ snowflake

CWI PHD GRADUATE FOUNDED RECORD IPO COMPANY SNOWFLAKE

Publication date: 18-09-2020

Data-warehouse company Snowflake went public this week, reaching an extraordinary market value of $70.4 billion, the largest IPO for a software company ever. Snowflake offers cloud-based data warehousing, whose data storage and query engine contain techniques pioneered in CWI's Data Architectures group. One of the group's PhD graduates who developed that technology, Marcin Żukowski, is co-founder of Snowflake.

VectorWise ⇒ Databricks (Spark)

CWI RESEARCH COLLABORATION LEADS TO 100 MILLION EURO INVESTMENT IN AMSTERDAM

Publication date: 06-11-2019

CWI's research partner Databricks announced it will invest 100 million euros to expand its Amsterdam office. The company aims to grow to hundreds of R&D engineers in the next three years. Databricks settled in Amsterdam almost three years ago because of its research collaboration with CWI. The company announced the investment during the Spark + AI Summit 2019, a data science event with more than 2000 visitors just held in Amsterdam.

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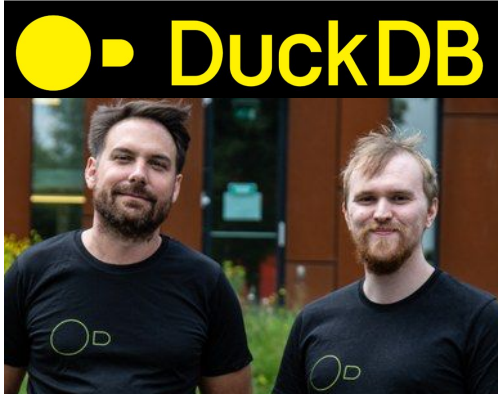
DuckDB

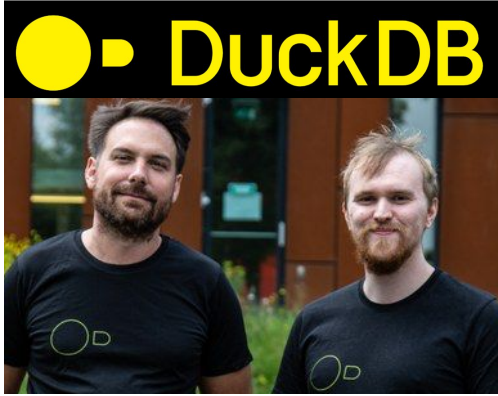
DuckDB: in-process analytics



- Created by Hannes Mühleisen and Mark Raasveldt
- Idea: **analytical SQL system as a linkable library**
- From research on data systems support for data science

https://duckdb.org/ https://shell.duckdb.org

DuckDB: in-process analytics



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- Idea: **analytical SQL system as a linkable library**
- From research on **data systems support for data science:**
  - why don't data scientists use database systems?
  - can transactions handle data science bulk-updates?
  - can we ensure reliability on cheap consumer hardware?
  - how to execute queries on compressed vector data?
  - scalable algorithms that respect changing/limited host resources?
  - …

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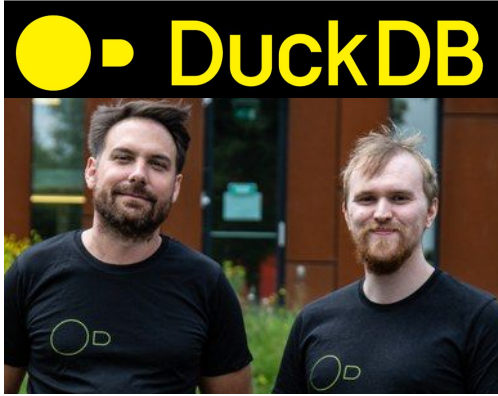
DuckDB: in-process analytics



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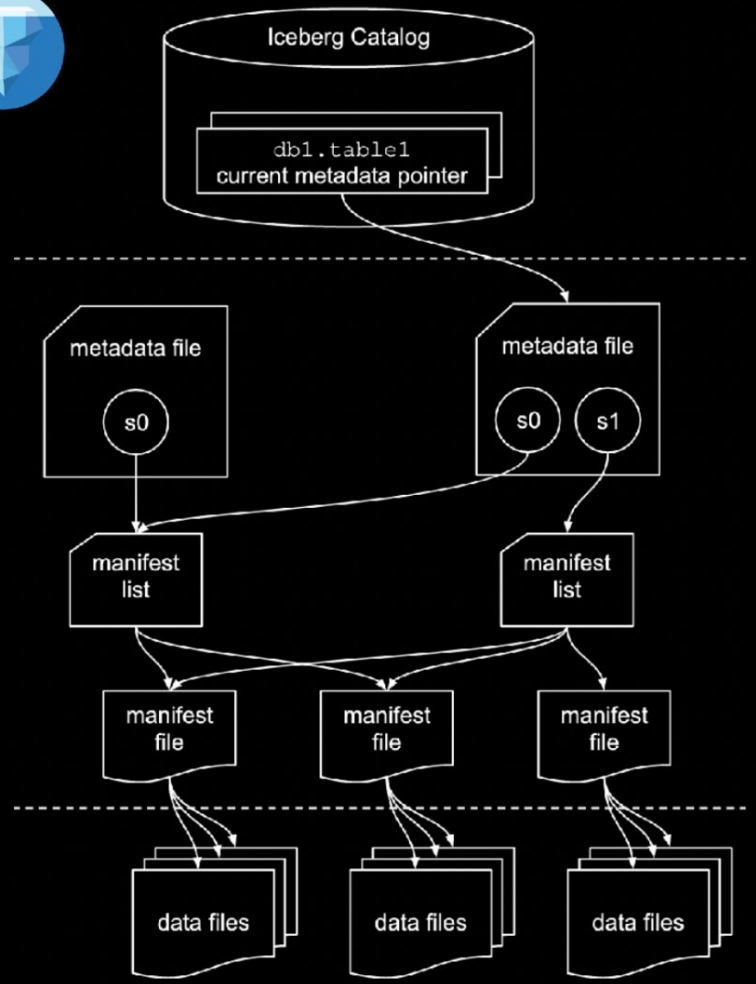
- Idea: **analytical SQL system as a linkable library**
- We foresee many applications beyond data science notebooks:
  - cloud analytics based on DuckDB
  - mobile apps that keep (most) data decentralized
  - edge analytics (e.g., intelligent streetlights, self-driving cars)
  - client-side (in-browser) query processing
  - computational storage (SQL in the cloud disks)

https://duckdb.org/ https://shell.duckdb.org

DuckLake


Iceberg Catalog
db1.table1
current metadata pointer
metadata file
s0
metadata file
s0
s1
manifest list
manifest list
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data files
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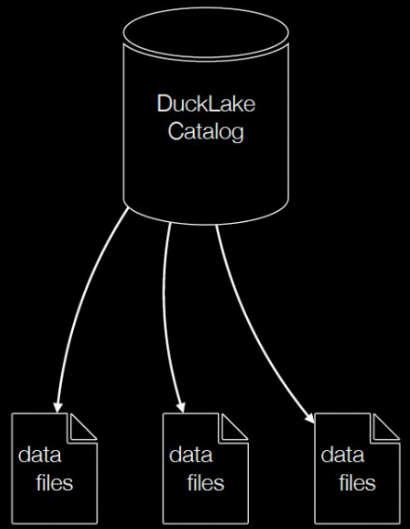
DuckLake Catalog
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MotherDuck

Incubating MotherDuck

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CWI SPIN-OFF DUCKDB LABS PARTNERS WITH MOTHERDUCK, WHO RAISES $47.5 MILLION

Publication date: 18-11-2022

CWI spin-off company DuckDB Labs helped create startup MotherDuck which aims to connect DuckDB to the cloud. MotherDuck sports some big names: its CEO is Jordan Tigani, founding engineer at Google's BigQuery, Google's fully managed data analysis platform. A big part of the $47.5 million funding comes from Andreessen Horowitz, a prominent venture capital firm, specialized in technology startups.

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MotherDuck

Incubating MotherDuck

News < Database startup MotherDuck raises $52 million to grow DuckDB-based platform

Database startup MotherDuck raises $52 million to grow DuckDB-based platform

Duck DB database technology was developed at CWI by the Database Architectures group.

Publication date 22 Sep 2023

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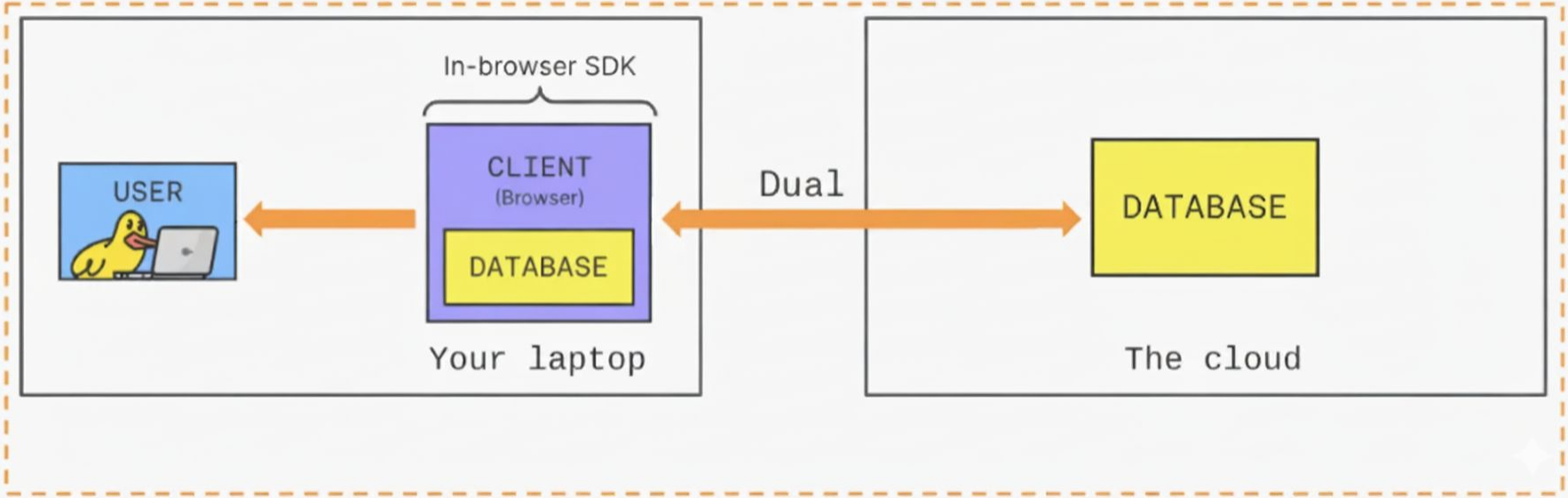
After raising $47.5 million last year, startup MotherDuck raised another $52 million, they announced this week. MotherDuck was founded in 2022 with help from CWI spin-off company DuckDB Labs, and is now ready to grow its DuckDB-based lightweight database platform.

The Seattle-based company, with a small office at CWI, now says to have nearly 2,000 users, including companies like ATM.com. This week MotherDuck removed the waitlist for its platform. Next step is bringing an upcoming release of DuckDB to the platform as well.

About DuckDB

MotherDuck

DuckDB in the Cloud and on your Laptop



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FastLanes

- **Problems**:
  - Parquet is GPU-unfriendly (because of snappy)
  - Compression ratio and decom speed can be improved
- **FastLanes**
  - new data-parallel encodings (SIMD friendly)
  - cascading encodings, expression encoding
- Future:
  - Idea: **learn** better ways of storing the data under the hood

Azim Afroozeh

https://github.com/cwida/FastLanes

```
INSTALL duckpgq FROM community;
LOAD duckpgq;
```

SQL/PGQ: Property Graph Query

- ISO started working with LDBC on graph query languages
- SQL:2023 will introduce PGQ:
  - Path-finding and pattern-matching

Daniel ten Wolde

```
CREATE PROPERTY GRAPH pg
  VERTEX TABLES (person PROPERTIES (personId, firstName))
  EDGE TABLES (knows SOURCE person DESTINATION person PROPERTIES(creationDate))
)
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SELECT gt.src, gt.dst
FROM GRAPH_TABLE (pg, MATCH (p1 IS person)-[k IS knows]*->(p2 IS person)
                      WHERE k1.creationDate = 2022/06/01
                      COLUMNS (p1.name AS src, p2.name AS dst)) gt
```
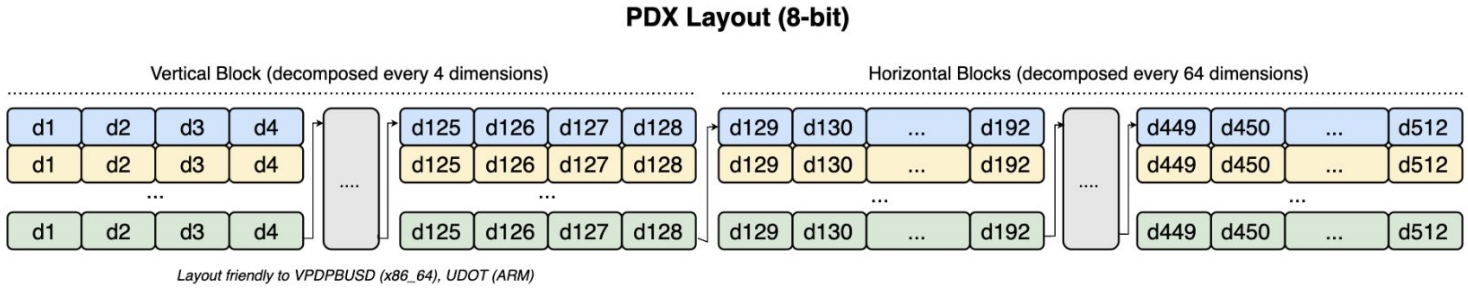

vector search



Leonardo Kuffo

- **PDX: store vectors as columns**
  - low memory footprint (<< HNSW)
  - cascading encodings, expression encoding
- Ongoing:
  - Low-cost continuous ingest
  - Filter pushdown
  - PDX as a DuckDB extension

https://github.com/cwida/PDX

CWI

Centrum Wiskunde & Informatica

Madelon Hulsebos | Azim Afroozeh | Effy Li | Ilaria Battiston | Daniel ten Wolde | Leonardo Kuffo | Lotte Felius | Paul Gross | Daniel Gomm | Cornelius Wolff | Omid Afroozeh

DuckDB

Hannes Muehleisen | Mark Raasveldt | Pedro Holanda | Gabor Szarnyas | Sam Ansmink | Tom Ebergen | Tania Gobatsch | Laurens Kuiper

Teamwork!

🙏

vectorwise

Marcin Zukowski | Sandor Heman | Alicja Luszczak | Juliusz Sompolski | Adrian Ionescu | Andrei Costea | Fabian Nagel | Tim Gubner

monetdb

Martin Kersten† | Sjoerd Mullender | Niels Nes | Stefan Manegold | Ying Zhang | Arjen de Vries | Lefteris Sidirourgos

databricks

Photon: A Fast Query Engine for Lakehouse Systems

Alexander Behm, Shoumik Palkar, Utkarsh Agarwal, Timothy Armstrong, David Cashman, Ankur Dave, Todd Greenstein, Shant Hovsepian, Ryan Johnson, Arvind Sai Krishnan, Paul Leventis, Ala Luszczak, Prashanth Menon, Mostafa Mokhtar, Gene Pang, Sameer Paranjpye, Greg Rahn, Bart Samwel, Tom van Bussel, Herman van Hovell, Maryann Xue, Reynold Xin, Matei Zaharia
photon-paper-authors@databricks.com
Databricks Inc.

ABSTRACT

Many organizations are shifting to a data management paradigm called the "Lakehouse," which implements the functionality of structured data warehouses on top of unstructured data lakes. This presents new challenges for query execution engines. The execu-

from SQL to machine learning. Traditionally, for the most demanding SQL workloads, enterprises have also moved a curated subset of their data into data warehouses to get high performance, governance and concurrency. However, this two-tier architecture is complex and expensive, as only a subset of data is available in the warehouse, and this data may be out of sync with the raw data due

MotherDuck

MotherDuck: DuckDB in the cloud and in the client

RJ Atwal, Peter Boncz, Ryan Boyd, Antony Courtney, Till Döhmen, Florian Gerlinghoff, Jeff Huang, Joseph Hwang, Raphael Hyde, Elena Felder, Jacob Lacouture, Yves Le Maout, Boaz Leskes, Yao Liu, Alex Monahan, Dan Perkins, Tino Tereshko, Jordan Tigani, Nick Ursa, Stephanie Wang, Yannick Welsch
firstname@motherduck.com

ABSTRACT

We describe and demo MotherDuck: a new service that connects

Reducing Cloud Footprint. An observation from analyzing tra of cloud data warehouses [2, 10] is that >95% of databases are <